Midterm Project Code Draft

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You should include all of your code for your midterm project here.

I've included a few chunks of code to help you get started, including:

- loading the tidyverse package, which includes dplyr, ggplot2, and some other useful packages
- an updated ggplot2 theme that approximates the Economist Graphic Detail figures. Feel free to change this or use a different theme!
- code to read in the CCC data
- a final chunk to save your figure

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.2     v readr     2.1.4
## v forcats   1.0.0     v stringr   1.5.0
## v ggplot2   3.4.2     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.0
## v purrr     1.0.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(RColorBrewer)
library(scales)
```

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##     discard

## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(grid)
library(ggtext)

theme_econ_andy <- function() {
  # Generate the colors for the chart procedurally with RColorBrewer
  palette <- brewer.pal("Greys", n=9)
  color.background = palette[1]
  color.grid.minor = palette[2]
```

```r
  color.grid.major = palette[3]
  color.axis.text = palette[8] #[6]
  color.axis.title = palette[8]
  color.legend.title = palette[8]
  color.title = palette[9]

  grey_8 <- palette[8]

  # Begin construction of chart
  theme_bw(base_size=9)  +

    theme(panel.background=element_rect(fill=color.background, color=color.background))  +
    theme(plot.background=element_rect(fill=color.background, color=color.background))  +
    theme(panel.border=element_rect(color=color.background))  +

    # Format the grid
    theme(panel.grid.major.y = element_line(color=palette[6], size=.04))  +
    ## NOTE: Some Economist figures don't include vertical grid lines.
    # To remove them, add + theme(panel.grid.major.x =  element_blank()) to
    # your ggplot plot
    theme(panel.grid.major.x = element_line(color=palette[6], size=.04))  +
    theme(panel.grid.minor=element_line(color=color.grid.minor, size=.03))  +
    theme(panel.grid.minor.y=element_blank())  +
    theme(axis.ticks.y =element_blank())  +
    theme(axis.ticks.x =element_line(color="black", size=.11))  +
    theme(axis.line.x = element_line(color="black", size=.11))  +
    # remove distance between plotting area and axis:
    theme(panel.border = element_blank()) +

    # Format the legend
    theme(legend.background = element_rect(fill=color.background))  +
    theme(legend.key = element_rect(fill=color.background))  +
    theme(legend.text = element_text(size=7,color=color.legend.title))  +
    theme(legend.title = element_text(size=8,color=color.legend.title))  +

    # Set title and axis labels, and format these and tick marks
    theme(plot.title=element_markdown(color=color.title, size=10, vjust=1.25))  +
    theme(axis.text.x=element_markdown(size=7,color=color.axis.text))  +
    theme(axis.text.y=element_markdown(size=7,color=color.axis.text))  +
    theme(axis.title.x=element_markdown(size=8,color=color.axis.title, vjust=0))  +
    theme(axis.title.y=element_markdown(size=8,color=color.axis.title, vjust=1.25))  +

    # facet stuff
    theme(strip.text = element_text(size=8,color=color.axis.text)) +
    theme(strip.background = element_rect(colour = color.background,
                                          fill = palette[2], size = 0.5)) +
    # Title fonts
    #theme(plot.title=element_text(family="OpenSans-CondensedBold", margin=margin(b=15)))  +
    #theme(plot.subtitle=element_text(family="OpenSans-CondensedLightItalic"))  +

    # Plot margins
    theme(plot.margin = unit(c(0.35, 0.2, 0.3, 0.35), "cm"))
}
```

```r
# Set the theme as the default, so it'll get added to all your figures.
theme_set(theme_econ_andy())
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
ccc <- read.csv("ccc_compiled_2023-10-11.csv")
```

Remember that you can combine separate figures into a big figure using patchwork. See https://patchwork.data-imaginist.com/ or the Lecture 4 notes for details.

```r
library(patchwork)
```

```r
ccc <- ccc %>%
  select(-starts_with("source"))

mi_data <- ccc %>%
  filter(state == "MI") %>%
  mutate(rj = str_detect(issues, "reproductive"))

ccc <- ccc %>%
  mutate(rj = str_detect(issues, "reproductive"))

ccc$month <- as.Date(paste(year(ccc$date), month(ccc$date),
                           "01", sep = "-"))
```

```r
us_rj <- ccc %>%
  filter(rj) %>%
  group_by(month, rj, valence) %>%
  summarize(count = n())
```

```
## `summarise()` has grouped output by 'month', 'rj'. You can override using the
## `.groups` argument.
```

```r
p1 <- ggplot(us_rj, aes(
    x = month,
    y = count,
    fill = factor(month)
  )) +
  geom_col(width = 25) +
  labs(x = "Month",
       y = "Number of Protests",
       title = "Protests in the US Overtime") +
  theme_classic() +
  scale_y_continuous(labels = scales::label_comma()) +
  theme(legend.position = "none") +
    annotate("rect",
```

```
          xmin = as.Date("2022-04-01"),
          xmax = as.Date("2023-01-01"),
          ymin = -Inf,
          ymax = Inf,
          alpha = 0.5,
          color = "grey20",
          fill = "grey20") +
      ggtext::geom_richtext(
        label = "Roe v. Wade Overturned",
        x = as.Date("2022-07-01"),
        y = 1100,
        color = "white",
         fill = "black",
        label.color = "black",
        size = 3.5)
```

```
ccc$year <- year(ccc$date)

us_data <- ccc %>%
     mutate(scaled_size_mean = scale(size_mean))

us_data <- ccc %>%
  filter(rj) %>%
  group_by(valence, year) %>%
  summarize(count = n(),
            size_mean = mean(size_mean, na.rm = TRUE)) %>%
  filter(!is.na(valence))
```

```
## `summarise()` has grouped output by 'valence'. You can override using the
## `.groups` argument.
```

```
us_data[is.na(us_data$size_mean), "size_mean"] <- 0

valence_lab <- c("Neither", "Left", "Right", NA)
names(valence_lab) <- c("0", "1", "2", NA)

p2 <- ggplot(us_data,
       aes(x = year,
           y = size_mean,
           fill = factor(valence),
           facet = valence
             )) +
  geom_col(position = "dodge") +
  labs(
    x = "Year",
    y = "Average Size of Protests",
    title = "Protests Separated by Valence in the US",
    fill = "Valence") +
  theme_classic() +
  theme(legend.position = "none") +
  scale_fill_discrete(labels = c("Neither", "Left", "Right", NA)) +
   facet_wrap(~valence,
              labeller = labeller(valence = valence_lab), scales = "free_y")
```

```r
mi_data$month <- as.Date(paste(year(mi_data$date), month(mi_data$date),
                               "01", sep = "-"))
mi_data <- mi_data %>%
     mutate(scaled_size_mean = scale(size_mean))

mi_rj <- mi_data %>%
  filter(rj) %>%
  group_by(month, locality) %>%
  summarize(count = n(),
            size_mean = mean(size_mean, na.rm = TRUE))
```

```
## `summarise()` has grouped output by 'month'. You can override using the
## `.groups` argument.
```

```r
mi_rj <- mi_rj %>%
  mutate(label = ifelse(!is.na(size_mean) & size_mean > 1000, as.character(locality), NA),
         fill = ifelse(!is.na(size_mean) & size_mean > 1000, "big", "small")) %>%
  mutate(fill = as.factor(fill))



p3 <- ggplot(mi_rj,
       aes(x = month,
           y = count,
           alpha = fill,
           label = label,
           color = label
       )) +
  geom_point(aes(size = size_mean)) +
  labs(x = "Month",
       y = "Number of Protests",
       title = "Uprisings in Michigan",
       size = "Average Size") +
  theme_classic() +
  scale_alpha_manual(values=c(.1, 1)) +
  #scale_color_manual(values=c("red", "grey")) +
  ggrepel::geom_text_repel() +
  guides(alpha = FALSE,
         color = FALSE) +
  theme(legend.position = "bottom") +
  theme(legend.box.background =
          element_rect(color = "black", size = 1)) +
  geom_vline(xintercept = as.Date("2022-06-024"), color = "black") +
  annotate("text",
         label = "Roe v. Wade Overturned",
         x = as.Date("2022-05-15"),
         y = 8,
         color = "black",
         size = 3.5)
```

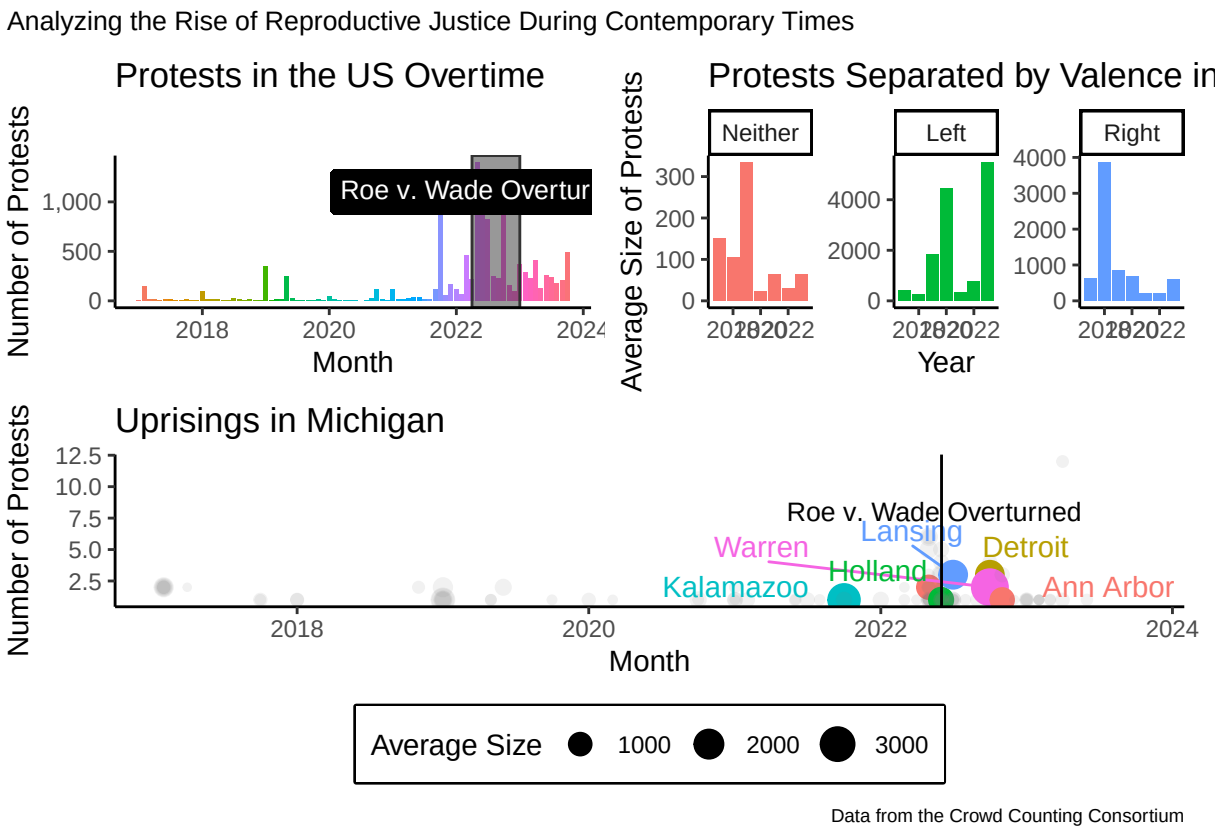
```
## Warning: The `<scale>` argument of `guides()` cannot be `FALSE`. Use "none" instead as
## of ggplot2 3.3.4.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
combo_plot <- (p1 + p2) / p3
combo_plot + plot_annotation(caption ="Data from the Crowd Counting Consortium",
                             title = "Analyzing the Rise of Reproductive Justice During Contemporary Ti
```

```
## Warning: Removed 209 rows containing missing values (`geom_point()`).

## Warning: Removed 313 rows containing missing values (`geom_text_repel()`).

## Warning: ggrepel: 1 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



Finally, you'll want to save your final plot to a pdf so you can load it into the project writeup Rmd document. To do that, you can use the `ggsave` function. You'll need to experiment which changing the size of the image to make it look good in your writeup.

```
ggsave(combo_plot, file="midterm_project_plot.pdf", width=10, height=6)
```

```
## Warning: Removed 209 rows containing missing values (`geom_point()`).

## Warning: Removed 313 rows containing missing values (`geom_text_repel()`).
```